

Bayesian Logistic Regression, Model Evidence, and Variable Selection

Student note

1 Purpose of this note

This note explains how Bayesian logistic regression can be used as a simple test case for understanding the normalizing constant estimated by Sequential Monte Carlo (SMC) samplers.

The main message is the following. In Bayesian inference, the posterior distribution has the form

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{p(y)},$$

where

$$p(y) = \int p(y | \theta)p(\theta) d\theta$$

is the normalizing constant. In ordinary posterior simulation, one often cares mainly about sampling from $p(\theta | y)$, so the normalizing constant may look like an annoying byproduct. But in model selection, this constant becomes central. If M denotes a model, then

$$p_M(y) = \int p_M(y | \theta_M)p_M(\theta_M) d\theta_M$$

is the *model evidence*, also called the *marginal likelihood*. It is a principled Bayesian score for comparing models.

The plan is:

- (a) understand Bayesian logistic regression for one fixed model;
- (b) understand the model evidence $p_M(y)$;
- (c) use SMC to estimate $p_M(y)$;
- (d) compare different logistic regression models by maximizing $p_M(y)$;
- (e) later, for larger model spaces, compare models by maximizing $p(M)p_M(y)$, where $p(M)$ is a prior over models.

For most of this note, we deliberately focus on the simpler rule

$$\widehat{M} = \arg \max_{M \in \mathcal{M}} p_M(y).$$

This is equivalent to assuming a uniform prior over the finite model list $\mathcal{M}$. This assumption is often too simple for large model spaces, but it is a good starting point for understanding what the normalizing constant means.

2 Binary regression and logistic regression

2.1 The data

Suppose we observe data

$$(x_1, y_1), \dots, (x_n, y_n),$$

where

$$x_i \in \mathbb{R}^p, \quad y_i \in \{0, 1\}.$$

The variable y_i is binary. For example:

- $y_i = 1$ if a patient has a disease, $y_i = 0$ otherwise;
- $y_i = 1$ if a tumour is malignant, $y_i = 0$ otherwise;
- $y_i = 1$ if a customer defaults, $y_i = 0$ otherwise;
- $y_i = 1$ if a passenger survives, $y_i = 0$ otherwise.

The vector x_i contains features or covariates. For example,

$$x_i = (\text{age, blood pressure, cholesterol, } \dots).$$

The purpose of binary regression is to model the conditional probability

$$\mathbb{P}(y_i = 1 \mid x_i).$$

2.2 Why not ordinary linear regression?

One might first try a linear model

$$\mathbb{P}(y_i = 1 \mid x_i) \approx \alpha + x_i^\top \beta.$$

But this is problematic, because $\alpha + x_i^\top \beta$ may be smaller than 0 or larger than 1. A probability must lie inside $[0, 1]$.

Logistic regression solves this by passing the linear predictor through a function mapping $\mathbb{R}$ to $(0, 1)$.

Define

$$\eta_i = \alpha + x_i^\top \beta.$$

The number η_i is called the *linear predictor*. It can be any real number. Logistic regression then sets

$$\mathbb{P}(y_i = 1 \mid x_i, \alpha, \beta) = \sigma(\eta_i),$$

where

$$\sigma(z) = \frac{1}{1 + \exp(-z)}.$$

The function σ is called the logistic function or sigmoid function.

Thus

$$\mathbb{P}(y_i = 1 \mid x_i, \alpha, \beta) = \frac{1}{1 + \exp(-\alpha - x_i^\top \beta)}.$$

2.3 The link function

Logistic regression can also be written using the logit link.

Let

$$\pi_i = \mathbb{P}(y_i = 1 \mid x_i, \alpha, \beta).$$

Then

$$\pi_i = \sigma(\alpha + x_i^\top \beta).$$

Equivalently,

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \alpha + x_i^\top \beta.$$

The transformation

$$\text{logit}(\pi) = \log \left(\frac{\pi}{1 - \pi} \right)$$

is called the logit function.

Therefore logistic regression says:

$$\text{logit}\{\mathbb{P}(y_i = 1 \mid x_i)\} = \alpha + x_i^\top \beta.$$

This is useful because the left-hand side can be any real number, while the probability itself is constrained to lie between 0 and 1.

The coefficients have an interpretation on the log-odds scale. If x_{ij} increases by one unit while the other variables are held fixed, then the log-odds change by β_j . Equivalently, the odds are multiplied by $\exp(\beta_j)$.

3 Bayesian logistic regression

3.1 Likelihood

For a fixed value of (α, β) , the model is

$$y_i \mid x_i, \alpha, \beta \sim \text{Bernoulli}(\pi_i),$$

where

$$\pi_i = \sigma(\alpha + x_i^\top \beta).$$

The probability mass function of y_i is

$$p(y_i \mid x_i, \alpha, \beta) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

Assuming conditional independence of observations given (α, β) , the likelihood is

$$p(y \mid X, \alpha, \beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i},$$

where X is the $n \times p$ design matrix.

The log-likelihood is

$$\ell(\alpha, \beta) = \sum_{i=1}^n [y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)].$$

A numerically stable equivalent expression is

$$\ell(\alpha, \beta) = \sum_{i=1}^n [y_i \eta_i - \log(1 + \exp(\eta_i))],$$

where

$$\eta_i = \alpha + x_i^\top \beta.$$

3.2 Prior

In Bayesian logistic regression, (α, β) is treated as unknown and random. We assign a prior distribution

$$p(\alpha, \beta).$$

A simple prior is

$$\alpha \sim N(0, s_\alpha^2), \quad \beta_j \sim N(0, s_\beta^2), \quad j = 1, \dots, p,$$

independently.

For example, after standardizing the columns of X , one may start with

$$\alpha \sim N(0, 3^2), \quad \beta_j \sim N(0, 1.5^2).$$

This is only a suggested default, not a universal rule.

The prior scale matters a lot. Since the model evidence averages the likelihood over the prior, changing the prior changes the model being compared. A very wide prior may put most of its mass on unrealistically large coefficients. A very narrow prior may force all effects to be almost zero. In model selection, this affects the evidence.

3.3 Why the intercept should be treated differently

The intercept α is different from the slope coefficients β_j .

If the columns of X are centered, then $x_i = 0$ corresponds roughly to an “average” observation. The intercept α then represents the baseline log-odds for an average observation:

$$\alpha = \text{logit}\{\mathbb{P}(y = 1 \mid x = 0)\}.$$

The slope β_j , on the other hand, represents the effect of changing one covariate. Therefore α and the β_j 's need not have the same prior scale.

A reasonable prior on the intercept should allow a wide range of baseline event probabilities. For example,

$$\alpha \sim N(0, 3^2)$$

puts substantial mass on baseline probabilities ranging from very small to very large. By contrast, the slopes can often have a somewhat smaller prior scale, especially after standardization.

3.4 Why column normalization is important

Before fitting Bayesian logistic regression, one should normally standardize the columns of X . For a continuous feature x_j , replace it by

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

and s_j is the sample standard deviation of column j .

This matters for three reasons.

First, it makes the prior comparable across coefficients. If age is measured in years and cholesterol is measured in mg/dl, then a one-unit change does not mean the same thing in the two columns. A prior such as

$$\beta_j \sim N(0, 1.5^2)$$

only makes sense uniformly across j if the columns have been put on comparable scales.

Second, it improves numerical stability for posterior simulation.

Third, it makes model comparison fairer. If the prior is not adjusted for the scale of each feature, then the evidence may reward or penalize variables partly because of arbitrary measurement units.

Binary variables can either be left as 0/1 or centered and scaled. For this project, the simplest consistent rule is:

- center and scale every non-intercept column;
- keep the intercept column separate and do not standardize it;
- use the same prior scale for all standardized slopes.

3.5 Posterior distribution

By Bayes' formula,

$$p(\alpha, \beta \mid y, X) = \frac{p(y \mid X, \alpha, \beta)p(\alpha, \beta)}{p(y \mid X)}.$$

The denominator is

$$p(y \mid X) = \int p(y \mid X, \alpha, \beta)p(\alpha, \beta) d\alpha d\beta.$$

This integral is the normalizing constant of the posterior distribution. It is also the model evidence for the logistic regression model.

In logistic regression, unlike simple conjugate Bayesian models, the posterior does not have a closed form. Therefore we use numerical methods such as MCMC, Laplace approximation, variational inference, or SMC.

For this project, SMC is especially useful because it can produce both:

1. approximate posterior samples from $p(\alpha, \beta \mid y, X)$;
2. an estimate of the normalizing constant $p(y \mid X)$.

4 Datasets for first experiments

At the beginning, do not do variable selection. Pick one dataset, preprocess it, fit one Bayesian logistic regression model, and inspect the posterior.

Some good beginner datasets are listed below.

Dataset	Size	Features	Comment
Breast Cancer Wisconsin Diagnostic	$n = 569$	$p = 30$	Clean continuous predictors
Statlog Heart	$n = 270$	$p = 13$	Small, binary outcome, no missing
Cleveland Heart Disease	$n = 303$	$p = 13$ commonly used	Some missing-value handling needed
Adult / Census Income	Large	mixed	Larger categorical example

4.1 Breast Cancer Wisconsin Diagnostic

This is a standard binary classification dataset. The goal is to classify a tumour as malignant or benign using numerical features computed from digitized images of fine needle aspirate samples.

This dataset is available in scikit-learn via

```
sklearn.datasets.load_breast_cancer.
```

It is also available from the UCI Machine Learning Repository.

This is a good first dataset because the features are numerical and the data are already clean. However, it has 30 features, so full enumeration of all variable subsets would involve

$$2^{30}$$

models, which is too many for a first model-selection experiment. For the first Bayesian logistic regression fit, this is fine. For variable selection, one should either use only a small subset of features or use a smarter search method.

4.2 Statlog Heart

This is a small heart disease classification dataset with 270 observations and 13 features. The target is absence or presence of heart disease.

It is a good candidate for this project because it is small and has no missing values. Some variables are continuous, some are binary, and some are categorical or ordered. To keep the first variable-selection experiment simple, we will use a reduced set of mostly continuous or binary variables.

4.3 Cleveland Heart Disease

The Cleveland Heart Disease dataset is another popular binary classification example. The original target takes values

$$0, 1, 2, 3, 4,$$

where 0 means no heart disease and positive values indicate presence of heart disease. For binary logistic regression, one commonly defines

$$y_i = \begin{cases} 0, & \text{if original target is 0,} \\ 1, & \text{if original target is 1, 2, 3, or 4.} \end{cases}$$

This dataset is slightly more annoying than Statlog Heart because some variables may require missing-value handling.

4.4 Adult / Census Income

The Adult dataset is a larger binary classification dataset where the goal is to predict whether income exceeds a threshold. It contains many categorical variables and requires one-hot encoding. This is useful later, but it is not ideal for the first evidence-based variable-selection experiment because the preprocessing choices already create many modelling decisions.

5 Variable selection

5.1 The problem

Suppose the full dataset contains p candidate predictors. A model $M \subseteq \{1, \dots, p\}$ selects a subset of these predictors.

For model M , write X_M for the design matrix containing only the columns selected by M . The logistic regression model is

$$y_i \mid x_{i,M}, \alpha_M, \beta_M \sim \text{Bernoulli}(\pi_{i,M}),$$

where

$$\pi_{i,M} = \sigma(\alpha_M + x_{i,M}^\top \beta_M).$$

The parameter dimension depends on M :

$$d_M = 1 + |M|,$$

where the 1 corresponds to the intercept.

The model evidence is

$$p_M(y \mid X_M) = \int p_M(y \mid X_M, \alpha_M, \beta_M) p_M(\alpha_M, \beta_M) d\alpha_M d\beta_M.$$

The evidence-based model-selection rule is

$$\widehat{M} = \arg \max_{M \in \mathcal{M}} p_M(y \mid X_M).$$

Equivalently, we may maximize the log evidence:

$$\widehat{M} = \arg \max_{M \in \mathcal{M}} \log p_M(y \mid X_M).$$

5.2 Why evidence is useful

The evidence is not just the best possible likelihood value. It is the average likelihood under the prior:

$$p_M(y) = \int p_M(y \mid \theta_M) p_M(\theta_M) d\theta_M.$$

This distinction is crucial.

A larger model usually has more flexibility. It can often achieve a higher maximum likelihood than a smaller model. But evidence does not merely ask whether the model can fit the data somewhere in parameter space. It asks whether the model expected the data before seeing them.

A large model can be penalized if it fits the data only in a very small region of its prior parameter space. This is the Bayesian Occam effect.

Here is the intuition. Suppose model M^+ is larger than model M . Imagine that M^+ contains many parameter values that do not help explain the data, and only a small part of its prior mass behaves like the smaller model M . Then the average likelihood under M^+ may be smaller than the average likelihood under M .

In an idealized case, suppose the larger model has parameter space S^+ , the smaller model corresponds to a subset $S \subseteq S^+$, and the likelihood is zero outside S . Suppose also that, conditional on $\theta \in S$, the prior under M^+ agrees with the prior under M . Then

$$p_{M^+}(y) = p_{M^+}(\theta \in S) p_M(y) \leq p_M(y).$$

So the larger redundant model is penalized because it wastes prior mass.

For ordinary variable selection with continuous priors, this argument is not literally exact, because the smaller model often corresponds to a lower dimensional subspace such as $\beta_j = 0$, which has zero probability under a continuous prior. Nevertheless, the intuition remains correct: if the data only support the extra coefficient near zero, then the larger model pays a prior volume penalty.

5.3 Why we first maximize only $p_M(y)$

The fully Bayesian model posterior is

$$p(M | y) = \frac{p(M)p_M(y)}{\sum_{M' \in \mathcal{M}} p(M')p_{M'}(y)}.$$

For this note, we mostly ignore $p(M)$ and maximize $p_M(y)$. This is the same as using a uniform prior over the finite list of candidate models:

$$p(M) = \frac{1}{|\mathcal{M}|}.$$

This is acceptable for a small, carefully chosen model space. It is not acceptable as a universal default for large model spaces. For example, if all subsets of p variables are considered equally likely, then the induced prior on model size is

$$|M| \sim \text{Binomial}(p, 1/2),$$

which concentrates around $p/2$. For large p , this may place too much prior mass on moderately large models.

Therefore:

- for a first experiment, maximize $p_M(y)$;
- for larger model spaces, maximize $p(M)p_M(y)$;
- for full Bayesian model averaging, compute or sample from $p(M | y)$.

6 Quick reminder: SMC and normalizing constants

Suppose we want to sample from a target distribution

$$\pi(\theta) = \frac{\gamma(\theta)}{Z},$$

where

$$Z = \int \gamma(\theta) d\theta$$

is unknown.

In Bayesian inference,

$$\gamma(\theta) = p(y | \theta)p(\theta),$$

and therefore

$$Z = \int p(y | \theta)p(\theta) d\theta = p(y).$$

For Bayesian logistic regression, a standard SMC sampler uses a sequence of tempered targets

$$\gamma_t(\theta) = p(\theta)p(y | \theta)^{\phi_t},$$

where

$$0 = \phi_0 < \phi_1 < \dots < \phi_T = 1.$$

The corresponding normalized targets are

$$\pi_t(\theta) = \frac{\gamma_t(\theta)}{Z_t}.$$

At $t = 0$,

$$\gamma_0(\theta) = p(\theta),$$

so if $p(\theta)$ is normalized, then

$$Z_0 = 1.$$

At $t = T$,

$$\gamma_T(\theta) = p(\theta)p(y \mid \theta),$$

so

$$Z_T = p(y).$$

The ratio between successive normalizing constants is

$$\frac{Z_t}{Z_{t-1}} = \int \frac{\gamma_t(\theta)}{\gamma_{t-1}(\theta)} \pi_{t-1}(\theta) d\theta.$$

For the tempered sequence,

$$\frac{\gamma_t(\theta)}{\gamma_{t-1}(\theta)} = p(y \mid \theta)^{\phi_t - \phi_{t-1}}.$$

SMC estimates each ratio by an empirical average over particles approximately distributed according to π_{t-1} . Thus

$$\hat{Z}_T = \prod_{t=1}^T \left(\frac{Z_t}{Z_{t-1}} \right).$$

Equivalently,

$$\log \hat{Z}_T = \sum_{t=1}^T \log \left(\frac{Z_t}{Z_{t-1}} \right).$$

In the logistic regression model-selection problem, the same SMC calculation is repeated for each model M . The estimate

$$\hat{Z}_M$$

is an estimate of

$$p_M(y \mid X_M).$$

Then model selection can be performed by ranking

$$\log \hat{Z}_M.$$

7 Concrete project: variable selection on Statlog Heart

7.1 Dataset

Use the Statlog Heart dataset.

The original dataset has 13 predictors. To keep the first experiment small and clean, use the following 8 predictors:

1. age;
2. sex;
3. resting blood pressure;
4. serum cholesterol;

5. fasting blood sugar indicator;
6. maximum heart rate achieved;
7. exercise-induced angina indicator;
8. oldpeak.

Define the binary target as

$$y_i = \begin{cases} 0, & \text{absence of heart disease,} \\ 1, & \text{presence of heart disease.} \end{cases}$$

Let the resulting raw design matrix be

$$X^{\text{raw}} \in \mathbb{R}^{n \times 8}.$$

Standardize every non-intercept column:

$$x_{ij} = \frac{x_{ij}^{\text{raw}} - \bar{x}_j}{s_j}.$$

The intercept is not included in this standardization. It is handled separately.

7.2 Model space

Let

$$\mathcal{P} = \{1, \dots, 8\}$$

be the set of candidate predictors.

A model is a subset

$$M \subseteq \mathcal{P}.$$

The model space is

$$\mathcal{M} = \{M : M \subseteq \mathcal{P}\}.$$

There are

$$|\mathcal{M}| = 2^8 = 256$$

models. This is small enough to enumerate.

The intercept is always included. Therefore the empty model $M = \emptyset$ is not a model with no parameters; it is the intercept-only model:

$$\text{logit}\{\mathbb{P}(y_i = 1)\} = \alpha.$$

For a model M , define

$$x_{i,M}$$

as the vector of selected predictors for observation i . The model is

$$y_i \mid x_{i,M}, \alpha_M, \beta_M \sim \text{Bernoulli}(\pi_{i,M}),$$

where

$$\pi_{i,M} = \sigma(\alpha_M + x_{i,M}^\top \beta_M).$$

7.3 Prior

Use the following prior for every model M :

$$\alpha_M \sim N(0, 3^2),$$

and, independently,

$$\beta_{M,j} \sim N(0, 1.5^2), \quad j = 1, \dots, |M|.$$

The prior density is therefore

$$p_M(\alpha_M, \beta_M) = \varphi(\alpha_M; 0, 3^2) \prod_{j=1}^{|M|} \varphi(\beta_{M,j}; 0, 1.5^2),$$

where $\varphi(\cdot; \mu, \sigma^2)$ denotes the Gaussian density with mean μ and variance σ^2 .

The same prior scale is used for every included standardized predictor. This is important. If different variables were on different numerical scales, then the same prior would not mean the same thing across variables.

7.4 Evidence for each model

For each $M \in \mathcal{M}$, the likelihood is

$$p_M(y \mid X_M, \alpha_M, \beta_M) = \prod_{i=1}^n \pi_{i,M}^{y_i} (1 - \pi_{i,M})^{1-y_i}.$$

The evidence is

$$p_M(y \mid X_M) = \int p_M(y \mid X_M, \alpha_M, \beta_M) p_M(\alpha_M, \beta_M) d\alpha_M d\beta_M.$$

Use an SMC sampler to estimate

$$\log p_M(y \mid X_M)$$

for all 256 models.

Then rank the models by

$$\log \hat{p}_M(y \mid X_M).$$

The selected model is

$$\hat{M} = \arg \max_{M \in \mathcal{M}} \log \hat{p}_M(y \mid X_M).$$

7.5 Recommended output table

For each model, produce a table containing:

Model	Variables	Dimension	Estimated log evidence
M_1	age, max-heart-rate	3	$\log \hat{p}_{M_1}(y)$
M_2	age, sex, oldpeak	4	$\log \hat{p}_{M_2}(y)$
...	...	...	...

Remember that the dimension includes the intercept:

$$d_M = 1 + |M|.$$

Also compute the evidence gap between the best model and each other model:

$$\Delta_M = \log \hat{p}_{\hat{M}}(y) - \log \hat{p}_M(y).$$

If Δ_M is small, then the data do not strongly distinguish the two models. If Δ_M is large, then the evidence strongly prefers the better model.

8 How to check whether variable selection worked

Evidence-based variable selection should not be accepted blindly. Here are several checks.

8.1 Check SMC stability

Run the SMC sampler several times for the same model. Since SMC is random,

$$\log \hat{p}_M(y)$$

will vary across runs.

For the top few models, repeat the SMC estimate many times and report:

- mean estimated log evidence;
- standard deviation of estimated log evidence;
- effective sample size behaviour along the tempering path;
- number of tempering steps selected by the adaptive schedule;
- acceptance rates of the MCMC mutation kernels.

If the difference between the best and second-best model is smaller than the Monte Carlo variability of the log-evidence estimator, then one should not claim that the best model has been decisively selected.

8.2 Train/test predictive performance

Split the data into training and test sets.

Use the training set to compute evidence and select a model:

$$\hat{M} = \arg \max_M p_M(y_{\text{train}}).$$

Then evaluate the selected model on the test set using posterior predictive probabilities:

$$p(y_{\text{test},i} = 1 \mid y_{\text{train}}) = \int \sigma(\alpha + x_{\text{test},i,\hat{M}}^\top \beta) p_{\hat{M}}(\alpha, \beta \mid y_{\text{train}}) d\alpha d\beta.$$

Useful metrics include:

- test log score;
- Brier score;
- classification accuracy after thresholding at 0.5;
- AUC;
- calibration plots.

The most Bayesian metric here is the test log score:

$$\sum_{i \in \text{test}} [y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i)],$$

where $\hat{\pi}_i$ is the posterior predictive probability.

Accuracy is useful but incomplete, because logistic regression is really predicting probabilities, not only hard class labels.

8.3 Compare with classical baselines

Compare the selected model with simple classical methods:

- maximum likelihood logistic regression using all variables;
- AIC or BIC model selection;
- cross-validated Lasso logistic regression;
- forward stepwise logistic regression.

The Bayesian evidence method need not agree exactly with these methods. But if it selects something wildly different, investigate why.

Possible explanations include:

- the prior scale is too wide or too narrow;
- the variables are highly correlated;
- the evidence differences are within SMC Monte Carlo error;
- the dataset is too small to identify a stable variable subset;
- the selected variables predict well but are not unique.

8.4 Prior sensitivity

Repeat the experiment with several slope prior scales, for example

$$\beta_j \sim N(0, 1^2), \quad \beta_j \sim N(0, 1.5^2), \quad \beta_j \sim N(0, 2.5^2).$$

If the selected model changes drastically when the prior scale changes slightly, then the conclusion is not robust.

This is not a bug. It is a feature of Bayesian model comparison: the prior is part of the model. However, it means one should report the prior sensitivity honestly.

8.5 Synthetic-data validation

Before trusting the method on real data, simulate data from a known sparse logistic regression model.

For example, generate

$$x_i \sim N(0, I_8)$$

and

$$y_i \sim \text{Bernoulli}[\sigma(\alpha + \beta_1 x_{i1} + \beta_3 x_{i3})].$$

Then apply the evidence-based variable-selection method to see whether it recovers variables 1 and 3.

Repeat this experiment many times for different sample sizes n , signal strengths, and correlations among predictors.

This helps answer basic questions:

- How much data is needed before evidence recovers the true model?
- What happens when two predictors are strongly correlated?
- What happens when the signal is weak?
- Does the SMC estimate of the evidence have low enough variance?

8.6 Stability under data perturbation

Repeat the analysis under different train/test splits or bootstrap resamples.

For each variable j , compute the fraction of runs in which it is selected:

$$\hat{q}_j = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{j \in \widehat{M}^{(b)}\}.$$

If the selected model changes dramatically from split to split, then it may be better to report a set of plausible models rather than a single selected model.

9 Larger model spaces

Enumeration works for $p = 8$ because

$$2^8 = 256.$$

But if $p = 30$, then

$$2^{30} \approx 1.07 \times 10^9,$$

which is far too many models to fit one by one.

For larger model spaces, two changes are needed:

1. introduce a prior $p(M)$ over models;
2. use a smarter search or sampling algorithm over model space.

The objective becomes

$$\widehat{M} = \arg \max_{M \in \mathcal{M}} p(M) p_M(y).$$

Equivalently,

$$\widehat{M} = \arg \max_{M \in \mathcal{M}} \{\log p(M) + \log p_M(y)\}.$$

9.1 Simple model priors

A simple prior is the independent inclusion prior:

$$p(M) = \rho^{|M|} (1 - \rho)^{p - |M|},$$

where $\rho \in (0, 1)$ is the prior probability that each variable is included.

If one expects sparse models, choose a small ρ , for example

$$\rho = \frac{k}{p},$$

where k is the prior expected model size.

Indeed,

$$\mathbb{E}[|M|] = p\rho.$$

Another simple prior is

$$p(M) \propto \exp(-\lambda |M|),$$

where $\lambda > 0$ penalizes larger models.

For interaction models, the model prior should usually enforce hierarchy. For example, if an interaction $x_j x_k$ is included, then the main effects x_j and x_k should also be included. This is called strong heredity.

9.2 Greedy forward selection

Start with the intercept-only model:

$$M_0 = \emptyset.$$

At step r , consider all models obtained by adding one variable:

$$M_r \cup \{j\}, \quad j \notin M_r.$$

Compute or estimate

$$\log p(M_r \cup \{j\}) + \log p_{M_r \cup \{j\}}(y)$$

for each candidate addition. Add the variable that gives the largest improvement.

Stop when no addition improves the score.

This is simple and often effective, but it can get stuck in local optima.

9.3 Forward–backward stepwise search

A more robust version alternates between adding and removing variables.

At each step, consider neighbours of the current model:

$$\mathcal{N}(M) = \{M \cup \{j\} : j \notin M\} \cup \{M \setminus \{j\} : j \in M\}.$$

Move to the neighbour with the largest score improvement:

$$S(M) = \log p(M) + \log p_M(y).$$

Stop when no neighbouring model improves the score.

One can also include swap moves:

$$M' = (M \setminus \{j\}) \cup \{k\}.$$

9.4 Stochastic search over model space

Instead of trying to deterministically find the best model, define a Markov chain on model space whose target distribution is

$$\pi(M \mid y) \propto p(M)p_M(y).$$

A simple proposal is:

- with some probability, add a random variable;
- with some probability, delete a random variable;
- with some probability, swap one included and one excluded variable.

If the current model is M and the proposed model is M' , accept the move with Metropolis–Hastings probability

$$a(M, M') = 1 \wedge \frac{p(M')p_{M'}(y)q(M \mid M')}{p(M)p_M(y)q(M' \mid M)}.$$

In practice, $p_M(y)$ is estimated by SMC. Therefore one should be careful: noisy evidence estimates can affect the behaviour of the Markov chain. A simple pragmatic approach is to cache evidence estimates and rerun SMC several times for promising models. A more formal approach is pseudo-marginal MCMC, but that is beyond this introductory note.

9.5 Screening followed by enumeration

For larger datasets, a practical approach is:

1. run a cheap screening method, such as Lasso logistic regression or marginal correlation screening;
2. keep only the top p_0 variables, for example $p_0 = 10$ or $p_0 = 15$;
3. enumerate all 2^{p_0} models in the reduced space;
4. compute evidence by SMC for those models.

This is not fully Bayesian, because the screening step uses the data. But it is a useful computational strategy for exploration.

9.6 Beam search

Beam search keeps several good partial models instead of only one.

For example, at model size r , keep the best B models of size r . Then expand each of them by adding one variable, score the candidates, and keep the best B models of size $r + 1$.

This is more expensive than greedy search but less expensive than full enumeration.

9.7 Evolutionary or genetic search

Another possibility is to represent a model by a binary vector

$$\gamma = (\gamma_1, \dots, \gamma_p) \in \{0, 1\}^p,$$

where

$$\gamma_j = 1$$

means variable j is included.

One can then use genetic algorithms, simulated annealing, or other black-box optimization methods to maximize

$$S(\gamma) = \log p(M_\gamma) + \log p_{M_\gamma}(y).$$

This is flexible but less statistically transparent than stepwise or MCMC search.

10 Suggested project structure

The student project can be organized as follows.

Stage 1: Bayesian logistic regression without variable selection

1. Choose one dataset.
2. Preprocess the target as $y_i \in \{0, 1\}$.
3. Standardize all non-intercept predictors.
4. Fit Bayesian logistic regression with all selected predictors.
5. Use SMC to sample from the posterior.
6. Report posterior means, credible intervals, and posterior predictive performance.
7. Report the SMC estimate of the normalizing constant.

Stage 2: Evidence-based variable selection

1. Choose a small set of $p = 8$ predictors.
2. Define $\mathcal{M}$ as all 2^8 subsets.
3. For each model M , run SMC and estimate $\log p_M(y)$.
4. Rank models by estimated log evidence.
5. Inspect the top models and evidence gaps.
6. Check SMC stability.
7. Evaluate posterior predictive performance on held-out data.

Stage 3: Larger model spaces

1. Introduce a model prior $p(M)$.
2. Use the score
$$S(M) = \log p(M) + \log p_M(y).$$
3. Try greedy search, forward–backward search, or stochastic search.
4. Compare the selected model with Lasso logistic regression and BIC.
5. Study prior sensitivity.

11 Main lessons

The normalizing constant is not just a numerical detail. In Bayesian model comparison, it is the model evidence:

$$p_M(y) = \int p_M(y \mid \theta_M) p_M(\theta_M) d\theta_M.$$

SMC samplers are useful because they can estimate this quantity while also sampling from the posterior.

For logistic regression, the evidence automatically balances fit and complexity. A larger model is not rewarded merely because it can fit the data somewhere in parameter space. It must fit the data well over a sufficiently large region of its prior parameter space.

However, evidence depends strongly on the prior. Therefore careful standardization, sensible prior scales, and prior sensitivity checks are essential.

For a first project, the cleanest setup is:

- binary logistic regression;
- standardized columns;
- intercept always included;
- Gaussian prior with separate intercept and slope scales;
- small model space, such as 2^8 subsets;
- model selection by maximizing $\log p_M(y)$;
- later extension to maximizing $\log p(M) + \log p_M(y)$.

References

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